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Republic of Somaliland

Somaliland National Examination Board

Form Four

MATHEMATICS PAPER ONE

Paper One (CORE MATHEMATICS)

JULY 2017

TIME 2 HOURS

Plus 10 minutes for reading through the paper

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INSTRUCTIONS TO CANDIDATES

This paper consists of 11 printed pages.
Count them now. Inform the invigilator if there are any pages missing.

PART 1: 30 Multiple Choice Questions 60 Marks

PART 2: 8 Structured Questions 40 Marks

TOTAL 100 Marks

- Answer ALL questions in Part 1 and 2.
- All answers must be written on this paper in the spaces provided immediately after each question. Only write on this exam paper.

Use this page for rough work. It will NOT be marked.

1. The first step in the process is to identify the problem or goal. This involves understanding the current situation and what needs to be achieved.

2. Once the problem is identified, the next step is to gather information. This can be done through research, interviews, or data analysis.

3. After gathering information, the next step is to analyze the data. This involves looking for patterns, trends, and insights that can help inform the decision-making process.

4. The next step is to develop a plan or strategy. This involves determining the best course of action to achieve the goal, taking into account the available resources and potential risks.

5. Once a plan is developed, the next step is to implement it. This involves putting the plan into action and monitoring progress along the way.

6. Finally, the last step is to evaluate the results. This involves assessing the outcomes of the process and determining whether the goal has been achieved.

PART ONE: Multiple Choice**(60 marks).**Choose the correct answer. Answer ALL the questions. Each question carries 2 marks.

- The angle of 620° lies on the quadrant of
 A) First quadrant B) Second quadrant
 C) Third quadrant D) Fourth quadrant
- The conversion of the $\frac{5\pi}{6}$ to degree is :
 A) 120° B) 135° C) 24° D) 150°
- The length of arc of radius 9 cm, that is subtending an angle of 40° at the centre is :
 A) 2π B). $\frac{3\pi}{2}$ C). $\frac{5\pi}{2}$ D). 6π
- The reference of $\sin 150^\circ$ is :
 A. 30° B. 60° C. 90° D. 45°
- The value of $\sin 135^\circ$ is :
 A $\frac{1}{\sqrt{2}}$ B. 3
 C 1 D. $\frac{\sqrt{3}}{2}$
- The identity $\sin^2 x + \cos^2 x$ is equal to :.
 A. -1 B. $\sqrt{3}$ C. 1 D. 0
- Lim $X \rightarrow 3$ $\left(\frac{2x + 4}{x + 2} \right)$
 A $\frac{7}{5}$ B. $\frac{6}{5}$
 C 5 D. 2

8. The differentiation of $y = 3x^3 - 1$ is :

- A. $9x^2$ B. $9x^2 - 1$ C. $3x^2$ D. $6x^2 - x$

9. The gradient of the curve $y = 4x^2 + 2$, at $x = 1$:

- A. 8 B. 4 C. -16 D. $\frac{1}{2}$

10. The derivative of $f(x) = 5 \sin 2x$ is :

- A. $-10 \cos 2x$ B. $10 \cos 2x$ C. $-5 \sin 2x$ D. $5 \cos 2x$

11. The integration of $f(x^2 + x - 4)dx$ is:

- A. $\frac{x^3}{2} + x^2 - 4$ B. $\frac{x^4}{3} + \frac{x^3}{3} + C$
B. $\frac{x^4}{4} + \frac{x^2}{2} - 4x + c$ D. $\frac{x^3}{3} + \frac{x^4}{2} - 4 + C$

12. The mode is :

- A. The middle value of the items B. The maximum frequency of the items
C. The largest value of the items D. The minimum value of the items

13. The range of the following items: 12, 17, 10, 29, 3, 6, 21 is :

- A. 28 B. 26 C. 30 D. 12

14. The 90th percentile of 25, 27, 30, 38, 40 is :

- A. 40 B. 38 C. 30 D. 27

15. 10% of 445 is :

- A. 4.045 B. 4.45 C. 0.445 D. 44.5

16. The value of the definite integral : $\int^1 (3x - 1)dx$ is:

- A. $\frac{1}{2}$ B. $\frac{3}{2}$ C. 1 D. 0

17. The value of $\cos (60 + \pi)$ is:

- A. 0 B. $\frac{3}{\sqrt{2}}$ C. $-\frac{1}{2}$ D. $\frac{1}{\sqrt{2}}$

18. $\lim_{x \rightarrow \infty} x^3 + 5x$ is :

- A. ∞ B. $-\infty$ C. 5 D. x^3

19. i^{18} is equal to :

- A. $-i$ B. $+i$ C. 1 D. -1

20. $(4 - 3i) + (7 + 5i)$ is equal to :

- A. $11 + 8i$ B. $11 - 8i$ C. $11 - 2i$ D. $11 + 2i$

21. $(2 + 3i)(2 - 3i)$ is equal to :

- A. 13 B. -5 C. -13 D. 12

22. The solution of $x^2 + 1 = 0$ is :

- A. i B. $-i$ C. -1 D. 1

23. If $z = 3 + 4i$ the radius is :

- A. 1 B. 7 C. 5 D. -7

24. The argument of $Z = 1 + \sqrt{3}i$ is :

- A. 30° B. 90° C. 60° D. 45°

25. A box contains 4 white and 3 black balls. If one ball is drawn at random, the probability of getting a white ball is :

- A. $\frac{1}{7}$ B. $\frac{4}{7}$ C. $\frac{4}{3}$ D. $\frac{3}{7}$

26. If a die is rolled, the probability of getting an even number is :

- A. $\frac{1}{2}$ B. $\frac{4}{6}$ C. $\frac{1}{6}$ D. $\frac{2}{6}$

27. The value of ${}_8P_2$ is :

- A. 20 B. 30 C. 15 D. 10

28. The simplification of $\frac{8!}{7! \times 2!}$ is :

- A. 15 B. 8 C. 2 D. 4

29. The probability of having a wedding on Friday is :

- A. $\frac{1}{30}$ B. $\frac{2}{7}$ C. $\frac{1}{7}$ D. $\frac{1}{12}$

30. The second derivative of $y = x^4 - 2x^3$ is :

- A. $4x^2 - 6x^2$ B. $12x^2 - 8$ C. $12x^2 - 12x$ D. $6x^3 - 6x$

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Part Two : 8 Structured Questions

(40 marks)

1. The arc subtending an angle of 45° at the centre, find its length if its radius is 8 cm?

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2. Find the limits of the following :

a) $\lim_{x \rightarrow 0} \frac{5x^2}{6x^2}$

b) $\lim_{x \rightarrow \infty} \frac{3x^2}{x^4}$

[The page contains faint, illegible horizontal lines, likely representing a very low-quality scan or a blank page.]

3. Find the gradient of the function $f(x) = 2x^3 - 5$ at the points (2, 11)

a)

- b) Integrate $f(x^3 - x^2 - 2x - 9)dx$

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4. Find the standard deviation of the items 5, 6, 7, 8, 9

[illegible]

5. A die is rolled and coin is tossed. Find the probability of getting 4 on the die and head of the coin?

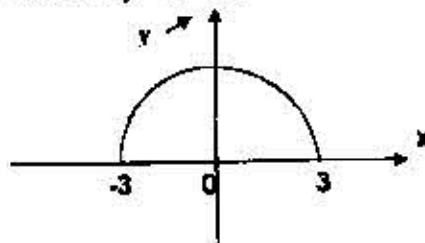
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6. Simplify the following

a) $\frac{1}{2+3i}$

b) Find the value of x and y that make the equation true $3x - 4yi = 18 - 24i$

7. Find the area under the curve
- $y = 9 - x^2$



8. Find
- ϕ
- of the following if
- $0 < \phi < 90^\circ$

a) $2 \sin \phi - 1 = 0$

b) $2 \cos \phi - \sqrt{2} = 0$